ITERATIVE METHODS FOR LINEAR SYSTEMS

$Ax = b \qquad A \in \mathbb{R}^{n\times n} \qquad b \in \mathbb{R}^n \qquad x_* = A^{-1}b$

Krylov subspace $\mathcal{K}_k(A,b) = \text{span}\{b, Ab, A^2b, \dots, A^{k-1}b\}$

Arnoldi algorithm: constructs $q_1, q_2, \dots, q_k$ orthonormal basis of $\mathcal{K}_k(A,b)$

↳ Arnoldi decomposition: $Q_k = \begin{bmatrix} q_1 & \cdots & q_k \end{bmatrix}$

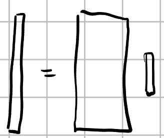
$$A \cdot Q_k = Q_{k+1} \cdot \underline{H_k} \qquad \underline{H_k} = $$  $\in \mathbb{R}^{(k+1)\times k}$

GMRES = generalized minimum residual

$\underset{x \in \mathcal{K}_k(A,b)}{\text{minimize}} \|Ax - b\| \rightsquigarrow x_k =$ the k-th approximation obtained by GMRES

if $x \in \mathcal{K}_k(A,b)$ then x can be written as a linear combination of the elements of the basis $q_1, \dots, q_k$

$\rightsquigarrow x = Q_k \cdot y \qquad y \in \mathbb{R}^k$



$$x_k = \underset{x \in \mathcal{K}_k(A,b)}{\text{argmin}} \|Ax - b\| = \underset{y \in \mathbb{R}^k}{\text{argmin}} \|AQ_k y - b\|$$

$$= \underset{y \in \mathbb{R}^k}{\text{argmin}} \|Q_{k+1}\underline{H_k} y - b\| = \underset{y \in \mathbb{R}^k}{\text{argmin}} \|Q_{k+1}\underline{H_k} y - Q_{k+1} e_1 \|b\|\|$$

Arnoldi relation

↳ $= Q_{k+1} \cdot e_1 \cdot \|b\|$

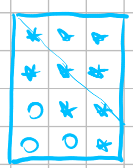
$$= \underset{y \in \mathbb{R}^k}{\text{argmin}} \|Q_{k+1}(\underbrace{\underline{H_k} y - e_1 \|b\|}_{\text{vector}})\|$$

$$= \underset{y \in \mathbb{R}^k}{\text{argmin}} \|\underline{H_k} y - e_1 \|b\|\|$$

$\|Q_{k+1} v\| = \|v\| \ \forall v$ because Q_{k+1} has orthonormal columns

↳ this is a least-squares problem of size $(k+1)\times k$ $\rightsquigarrow$ solving it costs $O(k^3)$

$k=3$



GMRES:
- $[Q_k, \underline{H_k}] = \text{Arnoldi}(A, b, k)$ — k mat-vec with A $+ O(nk^2)$
- $y_k \in \text{argmin} \|\underline{H_k} y - e_1\|b\|\|$ — $\leq O(k^3)$
- $x_k := Q_k \cdot y_k$ — $\rightsquigarrow O(nk)$

Some remarks:

1) the residual $\|Ax_k - b\|$ can be computed "for free" because it is equal to $\| \underline{H_k} y_k - e_1 \|b\| \|$.

/ without computing another mat-vec with A

2) Since $\underline{H_k}$ is upper Hessenberg, one can compute QR in $O(k^2)$

$$\begin{array}{l} \rightarrow \\ \rightarrow \\ \\ \\ \\ \end{array}\begin{bmatrix} * & * & * & * \\ * & * & * & * \\ & * & * & * \\ & & * & * \\ & & & * \end{bmatrix} \rightarrow \begin{matrix} * & * & * & * \\ 0 & * & * & * \\ & * & * & * \\ & & * & * \\ & & & * \end{matrix}$$

3) The QR decomposition of $\underline{H_k}$ can be "merged" with the Arnoldi loop (can be updated at each step).

if $A \in \mathbb{R}^{n \times n}$ symmetric $\rightsquigarrow$ Arnoldi becomes Lanczos

$\rightsquigarrow$ GMRES for symmetric matrices is called "MINRES".

Convergence analysis of GMRES [similar to analysis for CG]

Recall that $x \in \mathcal{K}_k(A,b) \iff x = p(A)\,b$ for some polynomial $p(t)$ of degree $\leq k-1$.

GMRES is solving
$$\min_{x \in \mathcal{K}_k(A,b)} \|Ax - b\| = \min_{\substack{x = p(A)b \\ \deg(p) \leq k-1}} \|Ax - b\|$$

$$= \min_{\substack{\text{poly } p \\ \text{with } \deg p \leq k-1}} \| A \cdot p(A) \cdot b - b \| = \min_{\substack{\text{poly } p \\ \text{with } \deg(p) \leq k-1}} \| (\underbrace{A\,p(A) - I}_{= \tilde{p}(A)})\, b \|$$

$\tilde{p}(t) = t \cdot p(t) - 1$

$\deg \tilde{p} \leq k$

[$p(t) = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \dots + \alpha_d t^d$

$\rightsquigarrow p(A) = \alpha_0 \underline{\underline{I}} + \alpha_1 A + \alpha_2 A^2 + \dots + \alpha_d A^d$]

if x_k is the approximation given by GMRES with $\mathcal{K}_k(A,b)$ then

$$\|Ax_k - b\| = \min_{\substack{\text{poly } \tilde{p} \\ \deg \tilde{p} \leq k \\ \tilde{p}(0) = -1}} \| \tilde{p}(A) \cdot b \|$$

if we have $A = VDV^{-1}$ $V \in \mathbb{R}^{n \times n}$ invertible, $D \in \mathbb{R}^{n \times n} = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots & \\ & & & \lambda_n \end{bmatrix}$

then $\tilde{p}(A) = V \cdot \begin{bmatrix} \tilde{p}(\lambda_1) & & \\ & \tilde{p}(\lambda_2) & \\ & & \ddots & \\ & & & \tilde{p}(\lambda_n) \end{bmatrix} V^{-1}$

$$\Rightarrow \|\tilde{p}(A)\cdot b\| \leq \|\tilde{p}(A)\| \cdot \|b\|$$

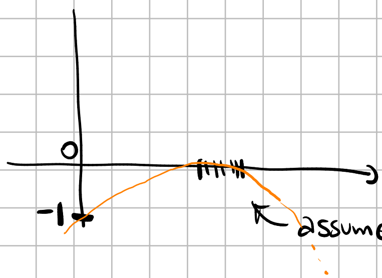
$$= \left\| V \cdot \begin{bmatrix} \tilde{p}(\lambda_1) & & \\ & \ddots & \\ & & \tilde{p}(\lambda_n) \end{bmatrix} V^{-1} \right\| \cdot \|b\|$$

$$\leq \underbrace{\|V\| \cdot \|V^{-1}\|}_{\kappa(V)} \cdot \left\| \begin{bmatrix} \tilde{p}(\lambda_1) & & \\ & \ddots & \\ & & \tilde{p}(\lambda_n) \end{bmatrix} \right\| \cdot \|b\|$$

$$\leq \kappa(V) \cdot \|b\| \cdot \max_{\lambda \in \{\lambda_1, \dots, \lambda_n\}} |\tilde{p}(\lambda)|$$

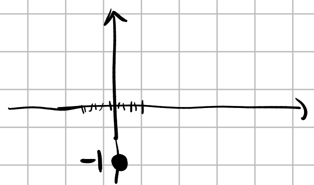
Theorem: if x_k is the k-th approximation of the solution of $Ax=b$ with GMRES, then

$$\|Ax_k - b\| \leq \left(\min_{\substack{\tilde{p} \text{ poly} \\ \deg \tilde{p} \leq k \\ \tilde{p}(0)=1}} \max_{\lambda \in \{\lambda_1, \dots, \lambda_n\}} |\tilde{p}(\lambda)| \right) \cdot \kappa(V) \cdot \|b\|$$



$\rightsquigarrow$ quick convergence :)

assume all eigenvalues of A are here



PRECONDITIONING

Convergence of GMRES depends on the eigenvalues of A

If they are "clustered", we're happy.

If not, we can modify the matrix:

$$\underset{(1)}{Ax = b} \quad \rightsquigarrow \quad \underset{(2)}{PAx = Pb} \quad \text{for some invertible matrix } P \in \mathbb{R}^{n\times n}$$

"left preconditioning"

The solution x_* of (1) is the same as the solution of (2), but GMRES applied to PA, Pb will give different iterates from GMRES applied to A, b.

If we choose P in a smart way, we could converge faster!

Non-examples: $P = I$ useless

$P = A^{-1}$ $\rightsquigarrow$ if you have it, then compute solution directly....

Examples: $P = \operatorname{diag}(A)^{-1} = \begin{bmatrix} a_{11}^{-1} & & & \\ & a_{22}^{-1} & & \\ & & \ddots & \\ & & & a_{nn}^{-1} \end{bmatrix}$

We wish:

- PA has clustered eigenvalues
- computing $P \cdot$ vectors can be done efficiently

$P = \begin{pmatrix} \text{upper triangular} \\ \text{part of } A \end{pmatrix}^{-1}$

$P =$ incomplete LU factorization

$\rightsquigarrow$ $P = \tilde{U}^{-1} \tilde{L}^{-1}$ where $\tilde{L}\tilde{U}$ is incomplete LU factorization

Right preconditioning: $Ax = b$ $\qquad \widetilde{AP}\, y = b$ $\to$ solve with GMRES(AP, b)

$x = Py$

Choosing a good preconditioner is an "art" and is very much problem-dependent!

Extra fact about Krylov methods:

one can extract approximations to the eigenvalues and eigenvectors of A from $[Q_k, \underline{H_k}] = \text{Arnoldi}(A, \text{random vector } b, k)$

$\underline{H_k} = \begin{bmatrix} H_k \\ 0\,0\,0\,0\,* \end{bmatrix}$ $\rightsquigarrow$ eigenvalues of H_k (μ_j, w_j) approximate some of the eigenvalues of A, and $Q_k \cdot w_j$ approximate the corresponding eigenvectors of A.

Eigs of H_k approximate the "external" eigenvalues of A.